
Advanced AI Dissertation

Design and Analysis of a Quantum Ensemble Framework with Coherent Aggregation

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A thesis submitted in part fulfilment of the degree of
Master of Science in Advanced Artificial Intelligence

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August 24, 2026

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List of Acronyms

CTG	Cardiotocography
CZ	controlled-Z gate
ECOC	Error-Correcting Output Codes
F1	macro-averaged F1 score
FYP	Final Year Project
HPC	High-Performance Computing
IBM	International Business Machines
JSON	JavaScript Object Notation
JSONL	JavaScript Object Notation Lines
NISQ	Noisy Intermediate-Scale Quantum
OOF	out-of-fold
QML	Quantum Machine Learning
QNN	Quantum Neural Network
QSVC	Quantum Support Vector Classifier
RBF	radial basis function
RX	X-axis rotation gate
RZ	Z-axis rotation gate
RZZ	two-qubit ZZ rotation gate
SEQSVM	Stacking Ensemble Quantum Support Vector Machine
SVC	Support Vector Classifier
SWAP	quantum state swap gate
UCI	University of California, Irvine
VQC	Variational Quantum Classifier

Glossary of Key Terms

Ansatz The chosen structure of a parameterised quantum circuit, including its encoding, trainable and entangling operations

Architecture Search A systematic evaluation of alternative model structures and design-variable combinations

Bagging An ensemble method that trains learners on different sampled subsets of the available training data and combines their predictions

Barren Plateau A region of the optimisation landscape in which gradients become too small to provide a useful training direction

Base Learner An individual classifier whose output or quantum representation contributes to an ensemble prediction

Binary Class Split A division of the original target classes into two groups that defines one binary classification task

Boosting An ensemble method that trains learners sequentially while increasing attention to samples or errors that earlier learners handled poorly

Circuit Depth The number of sequential layers of operations required to execute a quantum circuit

Circuit Width The number of qubits used by a quantum circuit

Class Imbalance A dataset condition in which some target classes contain substantially more samples than others

Codeword The binary pattern assigned to one target class across the columns of an error-correcting output-code matrix

Coherent Aggregation The combination of base-learner quantum representations within one joint circuit before final measurement

Data Encoding The process of mapping classical feature values to operations or states in a quantum circuit

Data Re-uploading The repeated introduction of input features between trainable circuit blocks

ECOC decomposition The conversion of a multiclass problem into several binary class splits whose predictions are recombined into a multiclass decision

Encoding Capacity The amount and structure of input information that a circuit configuration can represent through its available qubits and encoding operations

End-to-End Fine-Tuning A final training stage in which the complete composed model is unfrozen and its base and meta parameters are optimised jointly

Ensemble A model that combines the outputs or representations of several component learners to produce one final prediction

Entangling Layer A circuit layer containing multi-qubit operations that allow information to be shared between qubits

Feature Density The proportion of the available input features assigned to a circuit or ensemble learner

Feature Subspacing The assignment of different subsets of the original features to different learners

Features per Qubit The number of classical feature values encoded on each qubit during a data-upload block

Full-Register Aggregation A coherent meta-circuit design that can operate on every qubit belonging to the base-learner registers

Macro-averaged F1 Score The arithmetic mean of the class-specific F1 scores, giving every target class equal importance

Main-Wire Aggregation A coherent meta-circuit design that selects one representative qubit from each base-learner register

Measurement Boundary The point at which quantum information is converted into classical measurement outcomes

Measuring Aggregation An ensemble procedure that measures each base learner separately and combines the resulting classical outputs

Meta-Learner A model trained to combine the outputs or representations produced by the base learners

Out-of-Fold Training A stacking procedure in which meta-training representations are generated only from samples held out from the corresponding base-learner training fold

Parameterised Quantum Circuit A quantum circuit containing adjustable gate parameters that are optimised for a learning objective

Prediction Diversity The extent to which ensemble learners produce different predictions or errors on the same samples

QNode A PennyLane callable that represents and executes a quantum circuit on a selected quantum device

Qubit Register A designated group of qubits used to hold and process one component of a larger quantum circuit

Stacking An ensemble method in which a meta-learner is trained to combine the predictions produced by several base learners

Statevector Simulation Classical simulation that explicitly represents the complete complex-amplitude state of a quantum circuit

Stratified Split A data partition constructed to preserve approximately the same class proportions in each subset

Trainable Layer A circuit layer containing parameterised operations whose values are updated during optimisation

Validation-Weighted Decoding A direct aggregation rule that scales each base learner's contribution according to performance measured on validation data

Abstract

- Context: practical limitations of large Variational Quantum Classifiers (VQCs) and the motivation for a configurable ensemble of small quantum classifiers.
- Problem: quantum ensemble studies commonly fix the base circuit or aggregation procedure, limiting controlled analysis of how component design, ensemble construction and measurement affect the final model.
- Aim: implement QuantumECOC as a configurable multiclass quantum ensemble framework, analyse the VQC and ensemble design choices that determine its behaviour, and extend it through measurement-free coherent aggregation.
- Method: a custom parametric VQC implementation, the purpose-built QuantumECOC class, systematic component and ensemble searches, matched measured-aggregation comparisons and the CoherentECOC extension across multiclass tabular datasets.
- Evaluation: accuracy, macro-averaged F1 score (F1), class-level behaviour, convergence, training time and circuit-resource requirements.
- Principal findings: insert only after the final experiments and analysis are complete.
- Final conclusion: state what the main analysis reveals about QuantumECOC design and whether the CoherentECOC innovation is practically justified relative to measured aggregation.

Chapter 1: Introduction

1.1 Background and Motivation

Near-term Quantum Machine Learning (QML) is largely centred on hybrid variational models, in which [parameterised quantum circuits](#) are trained through classical optimisation [8]. Variational Quantum Classifiers (VQCs) apply this approach to classification by encoding input data, applying trainable circuit operations and measuring the resulting class probabilities. Their behaviour is strongly dependent on architecture, with closely related circuits able to produce substantially different predictive and optimisation results [16, 5].

The principal contribution of this dissertation is QuantumECOC, a custom quantum multiclass [ensemble](#) framework implemented through [Error-Correcting Output Codes \(ECOC\) decomposition](#). This reduces the task learned by each base circuit from multiclass classification to a [binary class split](#), allowing the ensemble to be constructed from small and computationally restricted VQCs. This shifts the principal opportunity for improving each learner towards the design of its circuit and encoded information rather than increasing its output complexity. The changing class partition assigned to each learner also introduces task diversity because every circuit models a different separation of the original classes. QuantumECOC manages these partitions, constructs the learners, assigns their features and samples, and combines their evidence into a multiclass prediction through one configurable and traceable implementation.

Ensemble learning can reduce dependence on one selected circuit, but its performance remains dependent on the information and decision capacity available to its learners [7, 10, 17]. The main empirical analysis therefore examines the design space of this framework at two connected levels. A configurable [ansatz](#) and VQC implementation allows the characteristics of the base circuits to be investigated systematically. The retained circuits are then evaluated as components of varying QuantumECOC ensemble architectures.

[Validation-weighted decoding](#) of the ECOC outputs provides the direct operating mode of QuantumECOC, with classical and measured quantum [stacking](#) implemented as supporting aggregation comparisons. The principal innovation is CoherentECOC, which implements [coherent aggregation](#) by retaining the pretrained base circuits as unmeasured components of a joint quantum circuit. It changes the [measurement boundary](#) as well as the combiner, allowing a trainable meta-circuit to operate on the joint base-register state before the final readout rather than learning from measured classical outputs.

1.2 Problem and Research Gap

Previous research shows that VQC performance is sensitive to circuit design and that ensembles of restricted quantum classifiers can offer an alternative to a single larger model [9, 10, 16]. These questions are generally investigated separately, however. Architecture studies concentrate on individual circuits, while ensemble studies frequently begin with a predetermined learner and aggregation rule. Consequently, there is limited controlled evidence connecting the design of a

configurable quantum component to its subsequent behaviour inside a multiclass ensemble.

Comparisons between aggregation methods can also be difficult to interpret when the learner pool, feature assignments or ensemble size change at the same time. An apparent improvement may arise from the revised ensemble rather than the method used to combine its outputs. Direct and stacked aggregation must therefore be compared using the same underlying learners and class decomposition.

Measured quantum stacking operates after each [base learner](#) has produced a classical output. Replacing the classical combiner with a quantum model does not remove this boundary because the measured values must first be encoded into a new circuit. Existing coherent proposals demonstrate that classifier contributions can instead be combined before measurement, but their construction differs from an ensemble of independently trained VQCs [23, 26]. It remains unclear whether a trainable coherent model built from such components can provide benefits sufficient to justify its additional cost.

This dissertation addresses the gap through an end-to-end configurable implementation centred on QuantumECOC. Its principal empirical analysis follows the custom VQC components into direct ensembles and determines how their design and composition affect the resulting system. Classical and measured quantum stacking provide controlled aggregation references, while the principal innovation, CoherentECOC, tests whether the same pretrained components can instead be combined before measurement. This structure separates the behaviour of the underlying ensemble from the effect of the aggregation mechanism.

1.3 Aim, Objectives and Research Questions

1.3.1 Overall Aim

The aim of this dissertation is to design and evaluate QuantumECOC as a configurable quantum multiclass ensemble framework, analyse how its VQC components and ensemble design affect predictive performance, stability and computational requirements, and extend the framework through the measurement-free CoherentECOC aggregation method.

1.3.2 Research Objectives

The research objectives are to:

1. Implement a configurable ansatz and VQC component that supports consistent circuit construction, training, prediction, resetting and resource tracing.
2. Implement QuantumECOC as a deterministic multiclass ensemble framework that manages ECOC decomposition, learner construction, feature and sample allocation, training, direct decoding and result recording.
3. Conduct a cross-dataset [architecture search](#) analysis of how VQC structure affects the performance, stability, information capacity and computational requirements of the available base components.
4. Extend this analysis to determine how learner count, architecture composition, feature avail-

ability and [prediction diversity](#) affect the complete QuantumECOC ensemble.

5. Implement classical and measured quantum stacking as matched aggregation comparisons against validation-weighted direct decoding.
6. Introduce and evaluate CoherentECOC, a measurement-free extension that combines pre-trained VQCs through a trainable meta-circuit before final measurement.

1.3.3 Research Questions

The dissertation addresses the following research questions:

1. Which VQC design characteristics provide the strongest balance between predictive performance, stability and computational requirements when used as configurable classifier components?
2. How are direct QuantumECOC results affected by learner count, base-architecture composition, information availability and prediction diversity?
3. Does classical or measured quantum stacking improve upon validation-weighted ECOC decoding when the underlying ensemble is held constant?
4. Can pretrained VQC base learners be combined through a trainable measurement-free meta-circuit, and does the resulting performance justify the additional circuit and optimisation cost?

1.4 Contributions and Scope

1.4.1 Contributions

The principal contribution of the dissertation is the design and implementation of QuantumECOC as a configurable quantum multiclass ensemble framework. Its principal innovation is CoherentECOC, which extends the framework so that pretrained base learners can be combined through a trainable quantum circuit without intermediate measurement. These contributions are supported by a large empirical analysis of the VQC and ensemble design choices that determine their behaviour.

The complete contribution therefore comprises:

- The purpose-built QuantumECOC class, which manages deterministic multiclass decomposition, custom VQC construction, learner-specific data allocation, training, validation-weighted decoding and result tracing.
- The CoherentECOC innovation, which transfers pretrained base ansätze and parameters into a joint circuit and applies a trainable meta-circuit before the only final measurement.
- The main cross-dataset analysis connecting VQC architecture and information capacity to direct ensemble performance, learner composition and realised prediction diversity.
- Matched direct, classical-stacked and measured-quantum aggregation comparisons that provide controlled reference conditions for the coherent extension.

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- The configurable ansatz and VQC implementation that supplies the reusable binary learner interface required by the ensemble framework.
 - An interactive dashboard for dynamically generating rankings, filtered design comparisons and repeated-run distributions across the complete analysis.

1.4.2 Scope and Delimitations

The empirical study focuses on supervised multiclass classification using small and moderate tabular datasets. Two-qubit VQCs are used as the principal ensemble learners so that the binary ECOC tasks remain computationally restricted and several learners can be combined. ECOC serves as practical multiclass decomposition scaffolding rather than a contribution to coding theory. The Stage III findings therefore characterise ensembles of small binary learners and should not be assumed to remain unchanged when the base-circuit width or task formulation is increased.

The broad model searches and coherent experiments are conducted through ideal [statevector simulation](#) because differentiable quantum simulation increases rapidly with circuit width and ensemble size. The dissertation evaluates observed performance–resource trade-offs and implementation feasibility, it does not attempt to establish a general quantum speed-up or universal superiority over classical learning.

1.5 Dissertation Structure

Chapter [2](#) reviews classical ensemble learning, VQC design, measured quantum ensembles, stacking and coherent aggregation. Chapter [3](#) describes the QuantumECOC implementation, the component and ensemble searches used for its main analysis, the measured aggregation comparisons and the CoherentECOC methodology. Chapter [4](#) presents the classical references, the VQC and direct ensemble findings, and the matched measured and coherent comparisons. Chapter [5](#) interprets these findings in relation to the research questions, previous work and computational trade-offs. Chapter [6](#) summarises the main contribution, innovation, analysis and opportunities for further development.

Chapter 2: Related Work

This chapter reviews the ideas needed to position the dissertation within classical ensemble learning and near-term QML. The discussion first introduces classical ensemble structure design, covering diversity, stacking and multiclass decomposition. It then examines the design and trainability of VQCs, before comparing quantum ensembles and various stacking approaches. The purpose is not to reproduce the full theoretical development of each method, but to identify the design choices, limitations and unresolved comparisons that directly motivate the experimental progression of the dissertation.

2.1 Classical Ensemble Foundations

2.1.1 Diversity, Bagging and Learned Aggregation

Dong et al. [7] describe ensemble learning as the combination of several predictive models to obtain a result that is more accurate or stable than the constituent learners considered separately. Common approaches include [bagging](#), [boosting](#), random forests, random-subspace methods and stacking. Their effectiveness does not follow automatically from increasing the number of learners. An ensemble benefits when its members are individually useful while making sufficiently different errors, creating a balance between predictive strength, diversity, stability and computational complexity.

Classical random-subspace methods create this variation by exposing learners to different feature subsets rather than requiring every model to process the complete input. Radojicic et al. [17] similarly use feature reduction to decrease the resources required by quantum ensembles while retaining variation between their learners. Architectural diversity alone is not sufficient, however, Pinto et al. [16] report that disagreement between base learner predictions can be more informative than either the individual [macro-averaged F1 score \(F1\)](#) or the structural difference between circuits. Diversity-aware approaches such as boosting therefore treat disagreement as an explicit optimisation variable alongside prediction loss and regularisation [12]. These findings motivate varying both the circuit structures and feature subsets of the base learners, while examining their realised prediction diversity rather than assuming that different architectures necessarily produce complementary errors.

2.1.2 Out-of-Fold Stacking and Model Selection

Stacking replaces a fixed prediction aggregation rule with a learned [meta-learner](#). Following the modified stacking approach examined by Demir and Dalkilic [6], the predictions of the base learners form a second-level feature representation from which a combiner model learns the final decision. This permits interactions between learner outputs to influence the result, rather than assigning every learner a fixed or independently calculated contribution.

The meta-training representation must be constructed from predictions for samples that were not used to fit the corresponding base learners. [out-of-fold \(OOF\) construction](#) satisfies this

requirement by training the base models on a subset of the training partition and predicting the held-out fold. Repeating the process across all folds produces meta-features without exposing the combiner to in-sample base predictions. Hybrid ensemble research has used the same principle to construct learned features before downstream quantum classification [28]. Demir and Dalkilic also show that feature-subspace generation, validation-driven learner selection and the choice of combiner can alter class-level outcomes [6]. Stacking must therefore be treated as a separate model design rather than a neutral replacement for direct aggregation, and its additional training cost must be considered alongside any predictive improvement.

2.1.3 ECOC

ECOC provide a method of decomposing a multiclass classification problem into a series of binary predictions. Following the formulation described by Nguyen et al. [14], each base learner divides the target classes into two groups and learns to distinguish between them. The grouping changes between learners, allowing each one to examine a different separation of the original classes.

The binary predictions are then combined to identify the target class whose expected pattern most closely matches the outputs of the learners. Since the final decision is based on several different class groupings, an individual incorrect prediction does not necessarily produce an incorrect multiclass result.

This decomposition is particularly useful for restricted quantum learners. Each circuit only needs to represent a binary decision boundary, reducing the output complexity required from an individual VQC and allowing circuit architecture, feature encoding and trainable capacity to become the principal learner-level design variables. The distinct class groupings also create diversity at the task level, even when the learners use identical ansätze, because their binary targets represent different views of the original multiclass problem.

2.2 Variational Quantum Classification

2.2.1 Near-Term VQC Workflow

Early QML research considered both the acceleration of expensive classical learning subroutines and the reformulation of probabilistic learning through quantum information processing [24]. Large theoretical speed-ups are generally associated with fault-tolerant algorithms and efficient access to quantum data, whereas current hardware is more commonly studied through hybrid variational models [8]. VQCs belong to this second group and use a parameterised quantum circuit as a trainable classifier within a classical optimisation loop.

The standard workflow described by Du et al. [8] consists of encoding classical features into a quantum state, applying parameterised transformations, measuring selected circuit outputs, calculating a loss and updating the parameters classically. The quantum circuit determines the accessible feature transformation, while the optimiser and loss provide the training procedure. Resch and Karpuzcu [20] emphasise that measurement exposes only limited information about a quantum state, so model outputs must be defined through probabilities or expectation values rather than direct access to the underlying amplitudes.

The practical value of a VQC cannot therefore be inferred only from its nominal qubit count or

Hilbert-space dimension. Biamonte et al. [4] identify data loading, state preparation and output reconstruction as important parts of the total computational cost, while Du et al. [8] distinguish circuit expressivity from trainability and generalisation. These limitations make controlled classical baselines, repeated optimisation runs and explicit resource reporting necessary when evaluating near-term quantum classifiers.

2.2.2 Encoding, Re-uploading and Ansatz Design

Data encoding maps classical features through parameterised gates before they can influence a VQC. Angle encoding associates individual feature values with rotation angles, while interaction-based encodings can introduce products or relationships between features. A single encoding layer restricts the frequencies available to the model. **Data re-uploading** addresses this restriction by interleaving repeated feature encodings with trainable operations, increasing the accessible function class without necessarily increasing the **circuit width** [8]. McFarthing et al. [13] relate repeated single-qubit feature rotations to a larger Fourier spectrum, although the additional **circuit depth** also increases the exposure to noise and optimisation difficulty.

The ansatz refers to how these encodings are arranged with trainable rotations and entangling gates. Its behaviour depends not only on which gates are present but on their order, distribution and connectivity. Salmenpera et al. [22] show that feature-map placement can materially change the resulting quantum model and can leave apparently present circuit operations functionally redundant. Entanglement can provide interactions across qubits, but broader connectivity and additional trainable entanglers increase both the parameter count and the two-qubit gate cost.

The number of interacting design choices creates what is often described as the ansatz lottery. Qubit count, encoding style, feature allocation, re-uploading, trainable-layer placement, entangling gate and connectivity can each take several forms, and these possibilities compound into a very large architecture space. Consequently, circuits with similar widths and depths can still differ substantially in their trainability and predictive performance. Pinto et al. [16] observe this variation between related VQC structures, while hardware-oriented research favours shallow templates and restricted entanglement to reduce optimisation and noise costs [9, 15]. There is therefore no single architecture that can be assumed to perform well across all datasets, supporting the empirical evaluation of these variables rather than relying on one fixed ansatz.

2.2.3 Trainability and Scaling

VQC optimisation is sensitive to the initial parameter distribution, circuit depth, data encoding and selected cost function [27, 9]. A central trainability problem is the **barren plateau**, in which gradients become too small to provide a useful optimisation direction. Cerezo et al. [5] relate this behaviour to circuit expressivity, global cost functions and increasing system size. The risk is therefore not resolved by making a circuit uniformly wider or deeper; the added representational capacity may become inaccessible to the optimiser.

Structured initialisation, local cost functions and restricted circuit designs are common responses [8]. Friedrich and Maziero [9] further demonstrate that replacing one larger Quantum Neural Network (QNN) with an ensemble of shallower components can increase gradient variance and reduce parameterisation complexity. Their findings connect circuit scale to the later ensemble question: distributing a task across trainable components may be preferable to adding all available capacity to one circuit.

2.3 Quantum Ensemble and Stacking Methods

2.3.1 Measured Quantum Ensembles

Measured aggregation executes component learners separately and combines the resulting classical outputs. Incudini et al. [10] apply bagging and AdaBoost to QNNs using different data-loading configurations, reporting reductions in qubit, parameter and gate requirements together with evaluations under noiseless simulation, backend-derived noise and real superconducting hardware. Friedrich and Maziero [9] approach modularity from the trainability perspective, while Park et al. [15] partition high-dimensional inputs across a multi-chip VQC ensemble. Although their implementations differ, each evaluates whether restricted quantum components can collectively replace a more demanding monolithic model.

Other work aims to adapt specific classical ensemble mechanisms, McFarthing et al. [13] construct bagging, boosting and logistic-regression ensembles from single-qubit variational classifiers. Song et al. [25] use quantum prediction probabilities as confidence information within AdaBoost.Q, experimentally evaluating the method on a superconducting processor. Bagging with a Quantum Support Vector Classifier (QSVC) has also produced application-level improvements in heart-disease classification [1]. Although these results show that measured quantum learners can benefit from classical combination, their different datasets, circuit models and evaluation conditions prevent the reported gains from establishing a general ensemble advantage.

Some proposals move more of the ensemble construction into the quantum procedure. Rathi and Kumar [19] use controlled operations to create transformed feature-label trajectories in superposition before applying a shared shallow classifier, while their unsupervised quantum bagging method uses QMeans learners and quantum sampling to improve resistance to corrupted labels [18]. Such methods expand the available design space but depend on specific state-preparation or memory-access assumptions and do not correspond directly to an ensemble of independently trained VQCs. Across these approaches, the predominant practical pattern remains the same.

2.3.2 Measured and Hybrid Quantum Stacking

Stacking extends measured ensembles by learning how the base outputs should be combined. Vasan et al. [27] combine multiple quantum classifier families in a stacking framework for industrial-control security, while Kang and Shin [11] integrate machine-learning, deep-learning and QNN outputs through a learned meta-model. These studies indicate that quantum predictions can contribute complementary information even when an individual QNN is not the strongest constituent model.

The location of the quantum component varies across hybrid stackers. Xiong et al. [28] use OOF classical stacking predictions as a reduced feature representation before downstream quantum classification. In the Stacking Ensemble Quantum Support Vector Machine (SEQSVM) approach, Akrom et al. [2] combine the outputs of classical boosting models and encode the resulting meta-features for a quantum support vector machine. Conversely, measured VQC or QNN outputs may be passed to a classical combiner [13, 27]. These arrangements evaluate whether a classical or quantum model can learn relationships between predictions.

2.3.3 Coherent and State-Preserving Aggregation

State-preserving ensembles attempt to perform the aggregation before the base information is converted into independent classical predictions. Schuld and Petruccione [23] propose representing multiple quantum classifiers through superposition, while Araujo and da Silva [3] examine an ensemble in which the contributions of trained classifiers are encoded and weighted within a quantum procedure. These approaches use quantum parallelism at the ensemble level, but require specialised representations of the classifier set rather than the simple concatenation of separately trained variational circuits.

Amplitude-based aggregation provides a related theoretical direction. Safina [21] applies quantum amplitude amplification to estimate the total class probability produced by a random-forest ensemble, with a theoretical quadratic reduction in the number of classifier queries. However, the simulated procedure loads classically calculated tree probabilities into quantum amplitudes rather than executing the trained base classifiers on quantum circuits.

Tolotti et al. [26] move closer to a coherent ensemble by using control registers to index homogeneous quantum classifiers. Validation measurements are used to estimate learner quality, after which the learned weights are encoded for coherent test-time combination. The method demonstrates that internal classifier contributions can be weighted within one quantum circuit, while also showing sensitivity to the accuracy of finite-shot weight estimates. More broadly, indexed data access, amplitude preparation, controlled classifier execution and post-selection can impose costs that are not reflected by the logical classifier count alone [23, 26].

2.4 Synthesis and Dissertation Gap

The literature establishes that VQC performance depends on circuit design and that ensembles can distribute a learning problem across smaller quantum components. Feature allocation, learner diversity and the aggregation rule may all influence the collective result [10, 13, 15]. Structural difference alone does not guarantee useful diversity, making the observed predictions of the learners as important as their nominal architectures [16, 17].

Existing studies examine separate parts of the resulting design space. These include quantum bagging and boosting, hybrid meta-learning, amplitude-based probability estimation and coherent classifier weighting [25, 2, 21, 26]. Their classifier families, data assignments and state-access assumptions differ, preventing their reported results from providing a controlled comparison of aggregation methods.

None of the reviewed approaches provides an end-to-end configurable implementation that constructs a deterministic ensemble of VQCs, systematically analyses its design space, and then retains the same ensemble across different aggregation methods. The dissertation aims to address this gap through the QuantumECOC framework and its CoherentECOC innovation, supported by controlled cross-dataset analysis and matched aggregation comparisons.

Chapter 3: Approach and Methodology

3.1 Research Design and Experimental Progression

The research was structured around the implementation and analysis of QuantumECOC. The analysis was structured as a staged empirical investigation in which model complexity increased while the number of candidate configurations was progressively reduced, allowing the main VQC and ensemble analysis to inform the more expensive aggregation experiments. This design was influenced by previous model-selection approaches that generated a large initial candidate pool before retaining a smaller set for later ensemble experiments [6]. It also responded to unresolved relationships between ensemble size, model diversity, predictive performance and training cost [7, 16].

Stage I established classical reference performance. Stages II and III formed the main empirical analysis: Stage II examined the configurable VQC components and Stage III determined how those components behaved inside the complete QuantumECOC framework. Stage IV implemented classical and measured quantum stacking as supporting aggregation comparisons. Stage V introduced the principal innovation, CoherentECOC, which reconstructed pretrained base learners within a joint circuit so that aggregation occurred before final measurement.

Each stage inherited the fixed datasets, interfaces and viable designs established previously, introducing only the variables required for the next experimental question. Candidate advancement followed the criteria in Section 3.9.3. The Stacked and Coherent comparisons retained the same ECOC matrix, learner count, base architectures and feature assignments. This progression separated changes caused by the aggregation method from those caused by rebuilding or reconfiguring the underlying ensemble.

3.2 Implemented Framework Architecture

QuantumECOC formed the centre of the implementation. Given an ensemble configuration, it generated the multiclass decomposition, constructed and trained the required binary learners, allocated learner-specific data, decoded their outputs and retained their predictive and resource information. The supporting ansatz layer represented circuits as configurable sequences of encoding, trainable and entangling operations, while the VQC class converted these structures into reusable learners with a common interface for construction, training, probability prediction, resetting and tracing.

Validation-weighted ECOC decoding formed the direct operating mode of QuantumECOC. StackedECOC reused the same ensemble state but exposed the measured learner probabilities to either a classical or quantum meta-learner. CoherentECOC reused the learned ansätze and parameters while changing the interface between the base and meta levels: the learners were placed on disjoint registers, their intermediate measurements were removed and a trainable meta-circuit was appended before the final readout. The four evaluated aggregation conditions therefore arose from a common ensemble definition rather than independent implementations.

3.3 Experimental Setup

3.3.1 Implementation Framework

The experimental framework combined PennyLane and PyTorch to construct and train the quantum models. PennyLane was used to define the parameterised quantum circuits as differentiable [QNodes](#), which were integrated with PyTorch through `qml.qnn.TorchLayer`. This provided a common training procedure for the individual VQCs, ECOC ensembles and quantum stacking models. Scikit-learn was used for dataset preprocessing, data splitting, classical reference models, classical meta-learners and evaluation metrics.

The implementation was divided across three main components. `Ansatzes.py` defined the reusable circuit layers and generated complete ansatz structures from ordered configurations. `QuECOC.py` contained the shared VQC implementation and the direct, stacked and coherent ensemble models. `Exploration.py` generated the experimental configurations, assigned deterministic identifiers and executed the repeated training runs.

Structured outputs from the execution framework were consumed by a Streamlit analysis dashboard. The storage and reproducibility procedure is described in [Section 3.10](#), while the available rankings and comparisons are defined in [Section 3.9](#).

3.3.2 Dataset Selection

14 multiclass classification datasets were chosen from the University of California, Irvine (UCI) Machine Learning Repository. The datasets were selected to provide variation in sample count, feature dimensionality, class count, class balance and feature type. This allowed the models to be evaluated across a range of classification conditions rather than a collection of structurally similar problems.

Eight main datasets were used for the individual VQC architecture search: Iris, Balance Scale, Contraceptive Method Choice, Heart Disease, Obesity, Image Segmentation, Steel Plates and Yeast. This group included small and large datasets, numerical and categorical features, low and high class counts, and different levels of [class imbalance](#). The main datasets were used to identify architecture characteristics that performed consistently across different problems before the candidate set was reduced for the ensemble experiments.

Six stress datasets were introduced after the VQC architecture shortlisting: Waveform, Cardiotocography (CTG) 10 Classes, Wine Quality, Nursery, Optical Recognition and Letter Recognition. These datasets extended the evaluation through larger sample counts, greater feature dimensionality, higher class counts and more complex class distributions. In particular, Optical Recognition and Letter Recognition increased the number of target classes, while Nursery introduced a larger categorical dataset and Wine Quality provided a strongly imbalanced ordinal classification problem.

3.3.3 Preprocessing and Data Splits

Each dataset was manually inspected before modelling to identify the target variable, feature types, missing-value markers and dataset-specific formatting requirements. Numerical, ordinal and categorical variables were distinguished according to the accompanying UCI documentation. Invalid values and duplicated headers were removed where necessary, while categorical variables

Table 3.1: Characteristics of the datasets used in the experiments. Feature counts are reported after preprocessing.

Dataset	Samples	Features	Type	Classes	Class proportion (%)	
					Min.	Max.
Iris	150	4	Numeric	3	33.33	33.33
Balance Scale	625	4	Numeric	3	7.84	46.08
Contraceptive Method Choice	1,473	11	Mixed	3	22.61	42.70
Heart Disease	303	20	Mixed	5	4.29	54.13
Obesity	2,111	23	Mixed	7	12.88	16.63
Image Segmentation	2,310	18	Numeric	7	14.29	14.29
Steel Plates	1,941	27	Mixed	7	2.83	34.67
Yeast	1,484	8	Numeric	10	0.34	31.20
Waveform	5,000	21	Numeric	3	32.94	33.92
CTG 10 Classes	2,126	21	Numeric	10	2.49	27.23
Wine Quality	6,497	11	Numeric	7	0.08	43.65
Nursery	12,960	19	Categorical	4	2.55	33.33
Optical Recognition	5,620	62	Numeric	10	9.86	10.18
Letter Recognition	20,000	16	Numeric	26	3.67	4.07

converted into numerical mappings using One Hot Encoding.

A single [stratified 80:20 train-test split](#) was created for each dataset using a fixed random state, and preprocessing operations requiring fitted values, including imputation and scaling, were fitted using the training partition only. The fully processed datasets were saved and used across all the models and stages, ensuring that performance differences resulted from the model configurations rather than changes in the sampled data.

Validation and cross-validation partitions were derived exclusively from the training set. The test partition was not used for preprocessing, early stopping, learner weighting, meta-model training or candidate selection. It was accessed only after training to provide the final evaluation of each completed model.

3.3.4 Training Configuration

All trainable quantum models used mini-batch optimisation with the Adam optimiser and negative log-likelihood loss. Training was limited to a maximum of 200 epochs and used a validation subset derived from the training partition. The training was stopped early when the validation loss failed to improve for the defined patience interval, following an initial warm-up period. The model state associated with the lowest validation loss was restored before test evaluation. The learning rate, batch size and early-stopping settings were held constant within each experimental stage.

Each quantum configuration was trained multiple times to obtain a distribution of performance results. This provided a measure of the variation introduced by quantum-circuit initialisation and optimisation. The deterministic classical reference models were evaluated once for each dataset because repeating them with an unchanged configuration would produce the same result.

3.4 Stage I: Classical Reference Baselines

This stage was implemented using the `SVC` and `RandomForestClassifier` estimator classes. Their shared estimator interface provided a consistent training and prediction workflow while retaining the distinct kernel-based and tree-based behaviour of the two model families.

3.4.1 Models

Two classical classifiers were evaluated before introducing the quantum models: a Support Vector Classifier (SVC) with a radial basis function (RBF) kernel and a Random Forest containing 100 decision trees. Balanced class weighting was applied to both models to reduce bias towards majority classes, with a fixed random state of 2 used throughout the experiments.

The two model families provide distinct classical reference points. The SVC represents a kernel-based method capable of learning non-linear decision boundaries, while the Random Forest provides a tree-based ensemble reference. Their configurations remained fixed across all datasets so that the comparison reflected differences between model families rather than dataset-specific hyperparameter optimisation.

3.4.2 Evaluation and Role in the Progressive Search

The resulting scores established the reference performance available before circuit architecture and quantum ensemble variables were introduced. They were used as practical points of comparison for the later models rather than as capacity-matched claims about the relative computational power of classical and quantum learning.

3.5 Stage II: VQC Component Analysis

The custom VQC class provided the common implementation for every individual quantum classifier tested during this stage. This search formed the component-level part of the dissertation's main analysis, identifying how the information and trainable capacity available to the learners could affect the framework built from them. The class resolved the selected features and ansatz template, constructed the PennyLane QNode and PyTorch quantum layer, and converted measured basis-state probabilities into normalised class probabilities. A consistent `fit`, `predict_proba`, `predict` and `score` interface was retained across the architecture search, together with functions for resetting a model and exporting its configuration, training history and results. Optimisation and model restoration followed the shared procedure defined in Section 3.3.4.

3.5.1 Parametric VQC and Ansatz Representation

The individual VQC formed the initial quantum baseline and supplied the circuit structures used by the later ensemble models. Each candidate was defined by its qubit count, selected input features and ordered ansatz specification.

A custom class was built to represent each ansatz as an ordered sequence of layer names and their associated parameters. Layers could be dynamically added and individually configured, allowing for various different ansatz designs to be easily implemented without the need to implement a separate circuit class for every configuration. VQC-level properties, including feature selection, were stored separately from the circuit-layer definitions. It also calculated the number of trainable parameters and encoded features, checked feature coverage and validated that each layer consumed the expected number of parameters. Template configurations were designed, allowing common structures to be initialised consistently and traced throughout the experimental process.

The circuit output was obtained using minimum measurement, with $\lceil \log_2(C) \rceil$ wires measured for a problem containing C classes. The resulting computational-basis probabilities were assigned to the target classes using a fixed class-state mapping and normalised before calculating the loss or generating predictions. Measurement width remained fixed according to the class count and was not included as an architecture-search variable.

3.5.2 Architecture Variables

The architecture search was designed around characteristics identified in previous VQC research as influential to circuit performance and trainability. These included circuit width, depth, the relationship between width and depth, [encoding capacity](#), feature reuploading, trainable-parameter distribution and entangling structure. Increasing circuit size can expand representational capacity, but can also increase resource requirements and optimisation difficulty. These effects depend on both the number of qubits and the number and arrangement of circuit layers [\[10, 5, 8\]](#).

Circuit width was controlled through the number of qubits, while depth was varied through the number of encoding, [trainable layers](#) and [entangling layers](#). Crossing these variables allowed the search to compare different depth-to-width relationships, including narrower circuits with repeated processing and wider circuits with fewer operations per qubit. Entangling gates were applied between successive reupload blocks, while the `trainable_layers` sequence specified how many trainable X-axis rotation gate (RX) and Z-axis rotation gate (RZ) layers occurred before each entangling position. Uniform, increasing, decreasing, front-loaded and back-loaded arrangements distinguished circuits with similar overall sizes but different distributions of trainable parameters, since layer placement can affect both resource use and the contribution of individual operations [\[22\]](#).

The amount of available input information requested by a circuit was controlled through [feature density](#). A density of 1.0 requested the complete available feature set, while lower fractional values deliberately restricted the circuit to a proportion of that information. The separate `feats_per_qubit` parameter controlled the [features per qubit](#) during every reupload. Input data were encoded using either single-feature angle rotations or pairwise encoding. Angle encoding applied input-dependent RX and RZ rotations, while pairwise encoding used two-qubit ZZ rotation gate (RZZ) rotations parameterised by the product of two features. The number of data reuploads was also varied, allowing additional features or repeated information to be introduced without increasing circuit width. Features were distributed between reuploads using block or cyclic allocation [\[8, 13\]](#).

Entangling capability was varied through both the two-qubit gate and its interaction pattern. Fixed controlled-Z gate (CZ) gates and trainable RZZ rotations were evaluated using linear and parallel patterns. The linear pattern applied adjacent interactions sequentially, while the parallel pattern grouped non-overlapping adjacent interactions within the same entangling stage. This allowed the effect of entangling placement to be compared while retaining the same available two-qubit operations. For two-qubit circuits, both patterns produced the same connection, so duplicate operations were removed. All candidate architectures were adapted to the Heron-oriented gate family used by the project, comprising RX, RZ, CZ and RZZ operations.

3.5.3 Search Execution and Evaluation

The individual-circuit search evaluated each VQC as a complete multiclass classifier. This differs from the subsequent ensemble stages, in which each base VQC learns one binary class split produced by the ECOC decomposition. The individual search therefore identified architectures capable of learning the more complex multiclass patterns without assuming that their relative ordering would necessarily be preserved when they were retrained on binary ensemble tasks.

Every valid VQC configuration was evaluated on the eight primary datasets using the common preprocessing procedure and fixed stratified train–test splits. The complete search contained 89,856 configurations per dataset, producing 718,848 configuration–dataset results. Each configuration was trained five times, with the data and architecture held constant while the parameter-initialisation seed varied between runs, resulting in a total of 3,594,240 fitted models.

For every run, parameter count, qubit count and training time were recorded alongside the predictive results. This allowed the subsequent ranking to distinguish architectures with similar classification performance but different computational requirements.

Configurations were ranked separately within each dataset using their mean test macro-averaged F1 score across the five initialisations. These ranks were converted to percentiles and then averaged across the eight datasets, so that a higher average percentile indicated stronger and more consistent relative performance across datasets of different difficulty.

The primary performance analysis was restricted to the configurations for which feature density was 1.0. This full-density setting requested the total available feature set and could therefore be used to assess the full predictive strength of each architecture, subject to the circuit's attainable encoding capacity.

The remaining density settings were retained for a separate robustness analysis. Because these settings provided each circuit with only part of the available feature information, they were not treated as directly comparable measures of maximum predictive performance. Instead, they were used to identify architectures whose relative performance remained stable as the amount of available input information was reduced.

The search evaluated widths of two, four, six and eight qubits and did not extend beyond eight-qubit circuits. It therefore examined ansatz behaviour within a relatively small-circuit regime rather than the optimal performance attainable from quantum models.

3.5.4 VQC Ranking and Shortlisting

The cross-dataset ranking was examined alongside grouped architecture comparisons to identify characteristics associated with performance, stability and training cost. This included comparisons based on qubit count, feature density, encoding, reuploads, trainable-layer arrangement and entangling structure.

Several strong configurations were retained for the ensemble stage rather than selecting a single highest-ranked circuit. The shortlist preserved variation in architecture and included resource-efficient configurations that remained close to the leading predictive results. Consistently uncompetitive designs were excluded because carrying the full architecture search into the more expensive ensemble experiments would provide limited additional evidence.

3.6 Stage III: QuantumECOC Ensemble Analysis

The QuantumECOC class was the central multiclass implementation evaluated in this stage. Given an ensemble configuration, it generated a deterministic ECOC matrix, constructed the required VQC base learners and retained the mappings connecting target classes, code columns, circuit structures and learner-specific features. It also handled input scaling, decomposition of the multiclass labels into binary targets and deterministic stratified sampling for the individual learner training sets. Once trained, the class combined learner outputs through validation-weighted ECOC support, produced multiclass probabilities and predictions, and recorded the ensemble structure, learner-level results, training history and resource requirements.

The architectures retained from Stage II supplied the reusable base-component pool. Together, Stages II and III formed the main analysis of the dissertation. This stage moved from individual component behaviour to the complete QuantumECOC design space, examining how the information and decision capacity available to the system changed through learner count, component composition and feature allocation while the underlying multiclass decomposition and training interface remained consistent.

3.6.1 ECOC Decomposition and Base Learners

ECOC were used to decompose each multiclass classification problem into a collection of binary subtasks. Each target class was assigned a binary [codeword](#), while each column of the code matrix defined the class partition learned by one base classifier. This reduced the complexity of the task assigned to each circuit and made the design and information capacity of the small base VQCs the principal learner-level variables. The different class partitions also created diversity between the learners because each column represented a distinct binary view of the original problem. The code matrices were generated deterministically so that matched ensemble configurations used the same binary decomposition.

Each binary subtask was learned using a two-qubit VQC constructed from the architectures short-listed during the individual VQC search. The restricted circuit width reduced the cost of training each learner and allowed larger ensembles to be constructed without requiring an equally wide circuit for every classification task. ECOC was used as a practical multiclass decomposition method rather than as a novel contribution to coding theory [14].

3.6.2 Ensemble Variability

The ensemble search varied the number of base learners, their architecture composition and their feature allocation. Ensembles containing 6, 10 and 14 learners were evaluated, and the architecture composition was divided into homogeneous, structurally related and broadly diverse groups. Homogeneous ensembles used the same ansatz template across all learners, while the other groups combined increasing levels of variation from the shortlisted VQC architectures. This allowed architectural diversity to be examined separately from the predictive strength of the individual candidates.

Feature subsets were varied across the base learners through [feature subsampling](#) even when their circuit architectures were identical, providing a method of introducing diversity while reducing the amount of information processed by each circuit [17]. The search included 3 feature density options of 0.25 and 0.75 and a mixed configuration, and 3 features per qubit options of 3, 5 and a mixed configuration. The block and cyclic allocation strategies from Stage II were reused at

learner level, while angle, pairwise, CZ and RZZ operations were inherited from the shortlisted architectures. At two qubits, the linear and ring patterns produce the same unique connection, so only one interaction structure was used.

Some mid-performing parameter configurations were also retained because Pinto et al. [16] found that differences between learner errors can have a greater influence on ensemble performance than the individual scores alone.

3.6.3 Direct Aggregation and Evaluation

The final prediction was obtained using validation-weighted ECOC decoding. Each learner's output was scaled by its validation performance to produce class-support values, and the class receiving the greatest combined support was selected. Only validation information derived from the training partition contributed to these weights.

Beyond the shared predictive and stability measures, ensemble results were interpreted alongside average learner quality and realised prediction diversity. This distinguished improvements associated with stronger components from those arising through complementary errors or increased learner count.

3.7 Stage IV: Measured Aggregation Comparisons

The `StackedECOC` class provided two supporting aggregation comparisons for `QuantumECOC`. It exposed the base probability vectors required by an interchangeable classical or quantum meta-learner and supported both single-split and k -fold meta-training through a common prediction and result-recording interface. This stage tested whether a learned interpretation of the same measured learner outputs was more effective than validation-weighted ECOC decoding and established matched references for the coherent innovation.

3.7.1 Candidate Pool and Matched Comparison

The stacking models used the leading configurations evaluated during the direct ensemble stage. For each matched comparison, the ECOC matrix, VQC architectures, feature subsets and learner count remained fixed. The direct condition retained validation-weighted decoding, while the two stackers learned from the measured base probabilities; the combiner was therefore the only altered component.

3.7.2 Meta-Training Procedures

Both procedures separated the data used to train the meta-model from the samples encountered by the corresponding base ensemble. This prevented the combiner from being trained on unrealistically favourable in-sample base predictions.

Single-split training. The training partition was divided using a stratified 70:30 split. The complete base ensemble was trained using the 70% subset, after which its predictions for the unseen 30% subset were used as the meta-training representation. The measured probability vector produced by each binary VQC formed the meta-features, while the original multiclass labels supplied the meta-targets. This procedure required only one base-ensemble fit but provided the meta-learner with a smaller training set and left 30% of the available data outside the base training process.

k -fold training. Following the out-of-fold stacking procedure used by Demir et al. [6], the training partition was divided into k stratified folds. For each fold, a complete copy of the base ensemble was trained using the remaining $k - 1$ folds and evaluated on the held-out fold. The held-out probability vectors were combined until every training sample had been processed by learners that had not encountered it during training. The reported fold-based procedure used $k = 3$ to balance meta-training coverage with the cost of repeatedly fitting the quantum ensemble.

The resulting OOF feature matrix was paired with the original multiclass labels and used to train the selected classical or quantum meta-learner. Preprocessing and feature scaling were fitted independently within each fold using only the fold-specific base-training data. Once meta-training was complete, the base ensemble was retrained using the full training partition. Its probability outputs were then used to construct the test meta-features supplied to the trained combiner.

3.7.3 Classical Meta-Learner

Following the use of classical meta-learning to combine measured quantum classifiers by McFarthing et al. [13], the classical stacking condition supplied the base probability vectors to an RBF-kernel SVC. The model used $C = 1.0$ and `gamma="scale"`; probability estimation was enabled and `random_state=2` was set for reproducibility.

The SVC learned the final multiclass decision directly from the measured base-learner probabilities rather than reconstructing an ECOC codeword through a fixed decoding rule. Its non-linear kernel allowed relationships between learner confidence values to influence the final prediction, including interactions that could not be represented through independent validation weights.

3.7.4 Quantum Meta-Learner

The quantum stacking model replaced the SVC with a fixed 12-qubit VQC. The measured base-learner probabilities were treated as classical meta-features and re-encoded into this circuit.

The meta-learner template used a dense feature strategy within the ansatz. It contained three data reuploads using angle encoding and cyclic feature allocation. The successive reuploads contained one, two and three trainable layers respectively. CZ entanglement was enabled across all reuploads and trainable layers using a parallel entangling pattern.

The meta-VQC was trained using the same optimisation and early-stopping procedure applied to the other trainable quantum models. Its final measured probabilities were mapped to the original multiclass labels, allowing it to learn interactions between the base outputs through a further variational quantum circuit.

3.7.5 Evaluation

The analysis examined whether learned aggregation improved upon validation-weighted ECOC decoding, whether the quantum meta-learner provided any benefit over the classical SVC and whether either effect depended on the retained ensemble characteristics.

Training time was reported for the selected meta-training procedure. Single-split training included one base-ensemble fit and the corresponding meta-model training. The three-fold procedure included three fold-specific base-ensemble fits, construction of the complete OOF representation and final base retraining on the full training partition. Classical stacking additionally included SVC fitting, while quantum stacking included optimisation of the 12-qubit meta-VQC. Repeated-run variation in classical stacking resulted primarily from the quantum base learners, while quantum stacking also included variation introduced by the initialisation and optimisation of the quantum meta-learner.

3.8 Stage V: CoherentECOC Innovation

CoherentECOC was the principal methodological innovation built on top of QuantumECOC. It extended the direct ensemble while removing the intermediate measurement between the base learners and the meta-model. Each trained base VQC was assigned a disjoint [qubit register](#), its learned parameters were transferred into a combined circuit and a configurable meta ansatz was applied before the only final measurement. The implementation supported both [main-wire aggregation](#), using one representative wire from each learner, and [full-register aggregation](#), using every base qubit. It could rebuild the combined circuit while preserving learned weights and optimiser state. The model also supported unfreezing the complete circuit for final [end-to-end fine-tuning](#). The base and meta resource structures were recorded alongside the final results.

3.8.1 Circuit Construction

The coherent condition begins from a selected and trained QuantumECOC configuration. Each base learner retains its circuit design but its terminal measurement is removed. The resulting unitary base circuits are assigned to disjoint registers within one larger QNode. The input sample is processed by every learner using the same feature mappings applied during its independent training, producing a joint state containing all base representations in quantum superposition.

A trainable meta-layer is appended after the base-circuit block and the complete circuit is measured only at its final output. The meta-circuit contains trainable and entangling operations but does not re-encode the original features or the measured probabilities used by StackedECOC. It therefore operates on the states prepared by the pretrained components rather than on a newly constructed classical meta-feature vector.

This construction differs from measured quantum stacking at a precise boundary. Measured stacking evaluates each base QNode, forms a classical probability vector and encodes that vector into a separate meta-VQC. CoherentECOC instead composes the unitary base ansätze and the meta ansatz within one differentiable circuit. The intended mechanism is the availability of cross-register operations before measurement, although the larger joint width also increases simulation and optimisation cost [20, 5].

3.8.2 Meta-Circuit Variability

Two main parameters controlled how the meta-circuit interacted with the base registers. The main-wire design selected one representative qubit from every learner and applied the cross-learner operations to those wires. Its active meta width therefore grew with the number of learners. The full-register design allowed the meta-circuit to act across every base qubit, exposing more of each learner state while producing a wider and more expensive interaction space. Due to limited time and computational resources the main-wire construction was used as the principal condition.

3.8.3 Coherent Training Procedures

CoherentECOC supported the same single-split and k -fold separation used by the measured stacking class. In both cases, the base circuits were trained before the coherent meta-layer so that its parameters were learned using samples that were unseen by the corresponding base learners.

For k -fold training, the learned ansätze and parameters were transferred into the unmeasured circuit, and the coherent meta parameters were updated using the held-out fold. One shared meta-parameter state and optimiser state were preserved across folds, while the fold-specific base learners were replaced before the next iteration. The reported fold-based procedure used $k = 3$. After all held-out folds had contributed to meta-training, the base ensemble was retrained using the complete training set. The coherent circuit was rebuilt using these final base ansätze and parameters while retaining the learned meta parameters.

Once the initial training was complete, the full circuit could be unfrozen and fine-tuned end to end using the complete training partition, allowing the pretrained base representations and coherent aggregation layer to adapt jointly [9, 8].

3.8.4 Coherent Evaluation and Finalist Selection

Coherent variants were constructed only for a reduced set of strong direct and measured-ensemble candidates as the complete circuit width was the sum of the retained base-register widths. For every matched comparison, the dataset split, ECOC matrix, learner count, base architectures, feature assignments and independently trained base tasks were retained. The newly introduced variables were therefore the coherent ansatz pattern, meta-circuit structure, final measurement width and whether final end-to-end fine-tuning was applied.

The coherent results were compared with validation-weighted direct decoding, classical stacking and measured quantum stacking built from the corresponding ensemble definition. The evaluation used mean macro-F1, repeated-run variation, convergence and total training time, together with the complete circuit width, depth and trainable-parameter count. These measures were used to determine whether the additional information available before measurement produced a predictive or stability benefit sufficient to justify the joint circuit and its increased optimisation requirements. Cross-register gates provide the interaction being tested, but also create the principal resource cost of the coherent extension [20, 15].

3.9 Cross-Stage Evaluation and Ranking

A common evaluation framework was used across the classical references, individual VQCs, direct quantum ensembles and the three aggregation extensions. The accumulated results were analysed through the interactive dashboard, which dynamically generated overall and dataset-specific rankings, configurable comparisons and repeated-run distributions. Results could be filtered by model family, circuit architecture, ensemble composition, aggregation method, meta-learner and resource requirements.

3.9.1 Common Predictive and Stability Metrics

Following the imbalance-sensitive evaluation used in previous classification studies [6, 27], macro-averaged F1 score was treated as the primary predictive metric. This assigned equal importance to every class and prevented performance on majority classes from dominating the evaluation, accuracy and weighted-F1 were reported as secondary aggregate measures.

Each trainable quantum configuration was evaluated across repeated runs. Following the use of repeated observations to evaluate both performance and stability in VQC ensembles [17], the mean, minimum, maximum and standard deviation were calculated for each recorded metric. The individual run results were retained so that distributions and variation between initialisation seeds could be examined rather than reporting only the mean result.

Applying the same measures throughout the experimental progression allowed changes in performance to be followed as circuit architectures, ensemble compositions and aggregation methods were introduced.

3.9.2 Global Ranking

Each configuration was ranked by its mean macro-averaged F1 score within each dataset, with equal scores receiving the same rank. The resulting per-dataset ranks were then averaged across all evaluated datasets to produce the global ranking. Every dataset contributed equally regardless of its sample count, feature dimensionality or number of classes, preventing larger datasets from exerting greater influence over the final ordering.

The dashboard dynamically generated these rankings for the selected model families and experimental stages. Rankings could also be grouped by aggregation method, learner count, architecture composition, circuit variables and resource characteristics. This allowed the strongest complete configurations to be identified while also examining whether particular design choices performed consistently across multiple datasets.

Average rank was used as the principal cross-dataset comparison because absolute macro-averaged F1 score values were not directly comparable between datasets with different levels of difficulty. However, the ranking was interpreted alongside the original macro-averaged F1 score values and repeated-run variation. This prevented small differences in rank from being treated as meaningful when the underlying predictive results were similar or unstable.

3.9.3 Progressive Selection Criteria

Following the progressive model-selection approach used by Demir et al. [6], fewer candidates were advanced as the cost of the experiments increased. Advancement decisions used the cross-dataset ranking defined above, but were not determined by rank alone. Raw macro-averaged F1 score, repeated-run stability and resource requirements were examined alongside the ordering, and consistently uncompetitive configurations were excluded from the more expensive stages.

The retained pool included several circuit structures and feature densities rather than a single winner. This maintained coverage of distinct design characteristics and prevented later conclusions from depending on one architecture family. Lower-cost configurations were also advanced when their predictive results remained competitive.

Two forms of analysis were then applied. Controlled comparisons used identical base configurations to isolate the effect of validation-weighted, classical-stacked and measured-quantum aggregation. The stacked comparisons retained the same ensemble definition for a smaller feasible subset. Grouped analysis used the wider candidate pool to examine relationships involving learner count, architecture composition and feature allocation.

3.10 Reproducibility and Computational Scope

Fixed stratified data splits, preprocessing values, feature mappings and label mappings were retained for each dataset. Feature allocation, ECOC generation, fold construction and learner sampling used deterministic seeds, while the parameter-initialisation seed was recorded separately and varied between repeated quantum runs.

Each completed run was stored in JavaScript Object Notation Lines (JSONL) format using a deterministic configuration identifier. The stored information included the model configuration, selected features, ansatz structure, training history, timing information and evaluation results. The search spaces, ensemble assignments and shortlisted configurations were also retained, allowing the dashboard rankings and comparisons to be regenerated without repeating the experiments.

The experimental scope was bounded by the increasing cost of statevector simulation and repeated ensemble training. Circuit width increased the simulated state space. Single-split stacking required one base-ensemble fit before meta-training, whereas the three-fold procedure required three fold-specific fits followed by final base retraining on the complete training partition. These costs determined the candidate budgets used at the later stages and are treated as boundaries of the experimental design rather than as failed experiments.

Chapter 4: Evaluation and Results

4.1 Stage I: Classical Reference Performance

The classical stage evaluated two foundational models, an SVC and a Random Forest, across the 14 datasets. Their results establish the baseline reference levels that can be achieved with minimal manual customisation and computational cost.

Table 4.1: Classical reference performance on the fixed test partitions.

Dataset	SVC		Random Forest	
	macro-averaged F1 scores	Accuracy	macro-averaged F1 scores	Accuracy
Iris	0.97	0.97	0.97	0.97
Balance Scale	0.71	0.78	0.58	0.80
Contraceptive Method Choice	0.45	0.47	0.49	0.52
Heart Disease	0.23	0.49	0.26	0.57
Obesity	0.58	0.59	0.95	0.95
Image Segmentation	0.86	0.86	0.98	0.98
Steel Plates	0.29	0.38	0.81	0.79
Yeast	0.56	0.59	0.58	0.59
Waveform	0.86	0.86	0.84	0.84
CTG 10 Classes	0.43	0.41	0.84	0.88
Wine Quality	0.08	0.11	0.39	0.68
Nursery	0.99	0.99	0.95	0.98
Optical Recognition	0.99	0.99	0.98	0.98
Letter Recognition	0.93	0.93	0.97	0.97
Dataset mean	0.64	0.67	0.75	0.82

Using the stronger classical model for each dataset we can achieve an optimal classical average macro-averaged F1 score of 77%, however, the performance ranged from 26% on Heart Disease to 99% on Nursery. Random Forest provided the stronger overall reference, with an average macro-averaged F1 score of 75%, compared with 64% for the SVC. It performed better on nine datasets, while the SVC performed better on four and the models tied on Iris.

These results also provide insight into the practical difficulty of the datasets, giving context for whether later performance differences reflect the model configuration or the underlying classification task. Nursery, Optical Recognition, Iris, Image Segmentation, Letter Recognition and Obesity were the easiest datasets for the classical models, with the strongest macro-averaged F1 score ranging from 95% to 99%. Waveform, CTG 10 Classes and Steel Plates also produced scores above 80% under at least one model. These datasets consequently provide relatively high reference levels for the later quantum experiments.

Heart Disease was the most difficult dataset, with neither model exceeding a macro-averaged F1 score of 26%. Wine Quality, Contraceptive Method Choice and Yeast were also comparatively difficult, reaching maximum scores of 39%, 49% and 58%, respectively. Balance Scale occupied an intermediate position, with the stronger model achieving 71%.

The results do not show a consistent relationship between class count and empirical difficulty, with

results varying significantly across low and high class counts. The balance of the class distribution provides a more consistent signifier, with the lowest performances all occurring on datasets with highly imbalanced classes.

Performance was also dependent on the selected classical model. Random Forest exceeded the SVC by more than 30 percentage points on Steel Plates, CTG 10 Classes, Obesity and Wine Quality, whereas the SVC produced the stronger result on Balance Scale. The subsequent quantum models are therefore compared with the stronger classical result for each dataset rather than with one fixed classical model.

4.2 Stage II: VQC Component Analysis

This stage presents the component-level part of the dissertation's main analysis. Its purpose is to identify the VQC and ansatz characteristics that show the greatest potential for modelling complex multiclass patterns and for supplying informative learners to QuantumECOC. Rather than treating one configuration as universally optimal, the results examine how circuit width, trainable capacity, layer placement, reuploading and feature allocation interact to produce strong and stable performance across the evaluated datasets.

4.2.1 Predictive Performance Across Datasets

Figure 4.1 shows the overall distribution of the mean F1 across the configurations on each dataset. Iris produced the highest central performance, with many configurations achieving strong results. The majority of the datasets demonstrate particularly wide distributions, showing that, even though a large proportion of the search performed poorly, suitable architectures could perform well. Balance Scale and Contraceptive Method Choice produced intermediate distributions, while Obesity, Heart Disease, Steel Plates and Yeast contained a greater concentration of low-scoring configurations. These differences demonstrate that the ability of an ansatz to model the data depended strongly on the dataset and reinforce the need to examine the strongest configurations alongside the complete search distribution.

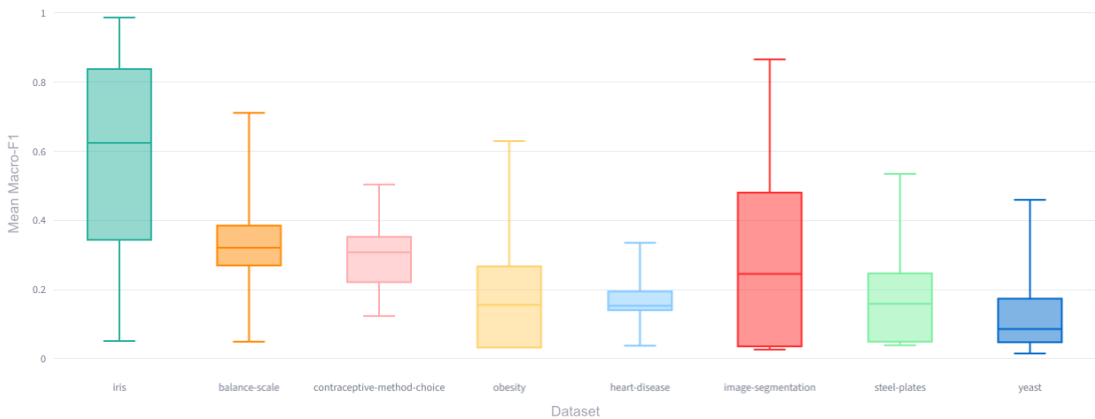


Figure 4.1: VQC performance distribution for the main datasets

Table 4.2 compares the five-run mean macro-averaged F1 score of the strongest VQC configuration on each dataset with the stronger of the two classical models. The best standalone VQC exceeded the classical reference on Iris, Contraceptive Method Choice and Heart Disease, and was effectively

level with it on Balance Scale. The largest positive difference occurred on Heart Disease, where mean macro-averaged F1 score increased from 26.00% to 33.53%. Iris produced the strongest absolute mean VQC result at 98.66%.

Table 4.2: Best VQC performance on each main dataset compared with the stronger classical reference.

Dataset	Mean VQC F1	Classical F1	Difference	Repeat SD	Qubits	Mean time (s)
Iris	0.986	0.97	0.017	0.018	2	5.60
Balance Scale	0.711	0.71	0.001	0.082	6	49.40
Contraceptive Method Choice	0.504	0.49	0.014	0.018	8	173.00
Heart Disease	0.335	0.26	0.075	0.049	4	8.40
Obesity	0.630	0.95	-0.320	0.059	8	519.40
Image Segmentation	0.866	0.98	-0.114	0.007	4	108.80
Steel Plates	0.535	0.81	-0.275	0.038	8	426.20
Yeast	0.459	0.58	-0.121	0.038	6	120.00
Dataset mean	0.628	0.719	-0.091	–	–	–

4.2.2 Trainable Capacity, Circuit Width and Training Time

Figure 4.2 relates average cross-dataset rank percentile to mean training time, with trainable parameter count represented by colour and circuit width shown in separate panels. The two-qubit configurations occupy a comparatively narrow and lower-ranked region, indicating limited capacity across the complete set of multiclass tasks, although the strong Iris result in Table 4.2 shows that this width can still be effective for an appropriate dataset. The largest width-related improvement occurs between two and four qubits, where configurations first reach the highest rank percentiles. The four-, six- and eight-qubit panels overlap substantially, demonstrating that additional width is not a strong indicator for performance. Configurations with more trainable parameters occur more frequently near the upper performance envelope, but also tend to require longer training times, reinforcing the importance of the performance–resource trade-off.

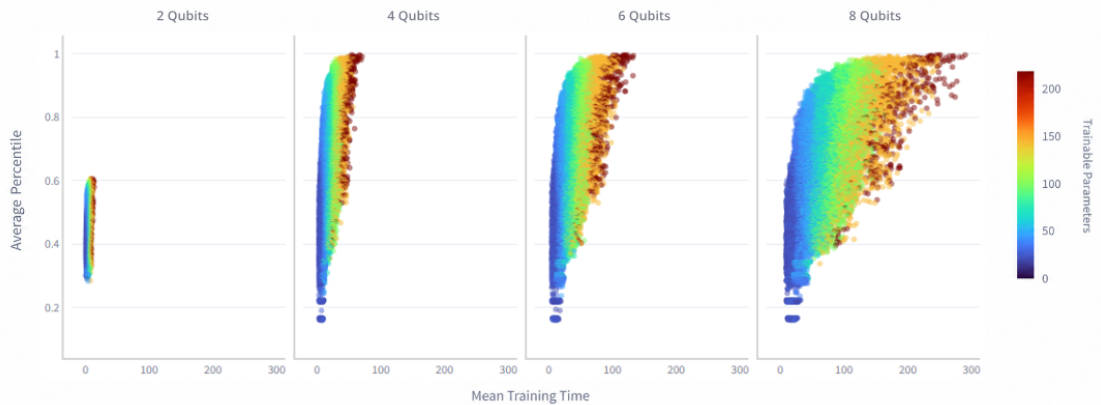


Figure 4.2: Mean Training time and Average Rank Percentile by Circuit Width and Trainable Parameter Count.

4.2.3 Reuploads and Trainable Layer Placement

The number and placement of trainable layers also affected performance. Each configuration is denoted by a list of integers representing the number of trainable layers in successive reupload

blocks. For example, [1, 2, 3] contains one trainable layer in the first block, two in the second and three in the third, giving six trainable layers in total.

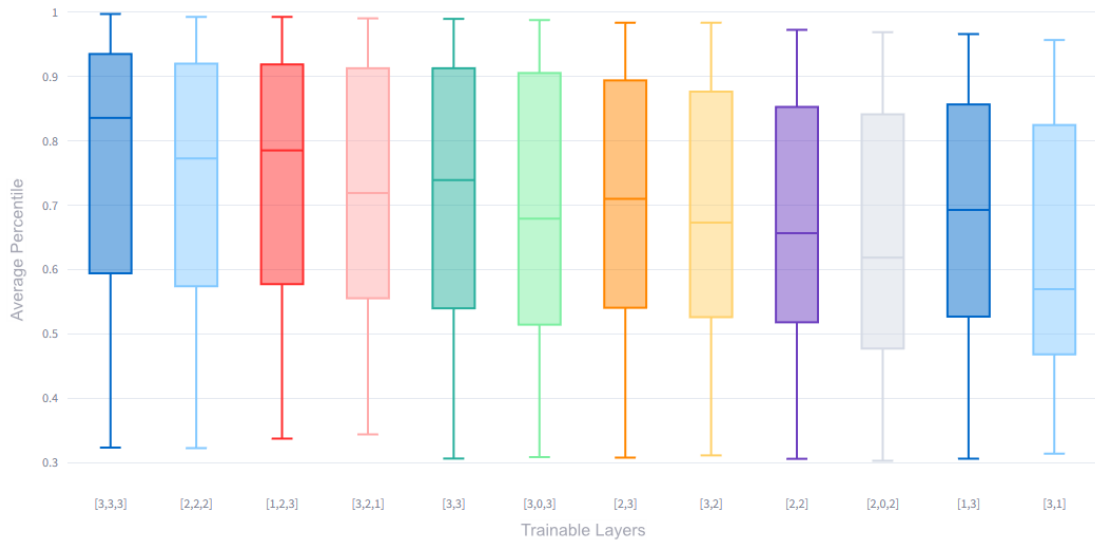


Figure 4.3: Average rank percentile by trainable-layer count and placement.

Among configurations with comparable trainable capacity, placing more layers later in the circuit generally produced a higher median rank. This pattern is visible in the matched comparisons [1, 2, 3] versus [3, 2, 1], [2, 3] versus [3, 2], and [1, 3] versus [3, 1]. Distributing six layers across three reupload blocks also performed better than concentrating them into two: both [2, 2, 2] and [1, 2, 3] exceeded [3, 3]. However, [3, 0, 3] shows that inserting an additional reupload block without corresponding trainable operations actually hinders performance. The ordering of the layers therefore matters alongside their total number.

4.2.4 Features per Qubit and Feature Strategy

The `feats_per_qubit` value determines how many features are encoded onto each qubit during every reupload. The `feature_strategy` parameter determines how the features are allocated to the qubits. The two strategies used in this search were block allocation, where features are encoded in contiguous blocks, and cyclic allocation, where features are encoded in a round-robin fashion. These two parameters work hand-in-hand to determine how much information is encoded into the circuit and how it is distributed across the qubits.

Figure 4.4 shows a strong interaction between the number of features per qubit and the allocation strategy. Block allocation remains comparatively robust at one and two features per qubit, whereas cyclic allocation produces markedly lower median ranks at these settings. At three features per qubit, the cyclic distribution improves substantially and reaches a median comparable to, and slightly above, the block strategy. Increasing the feature count therefore does not produce a uniform improvement independent of strategy, its main effect is to make cyclic allocation considerably more effective once sufficient total encoding capacity is available. The highest median is obtained by the three-feature cyclic configurations, although the overlapping distributions show that this combination is not universally superior.

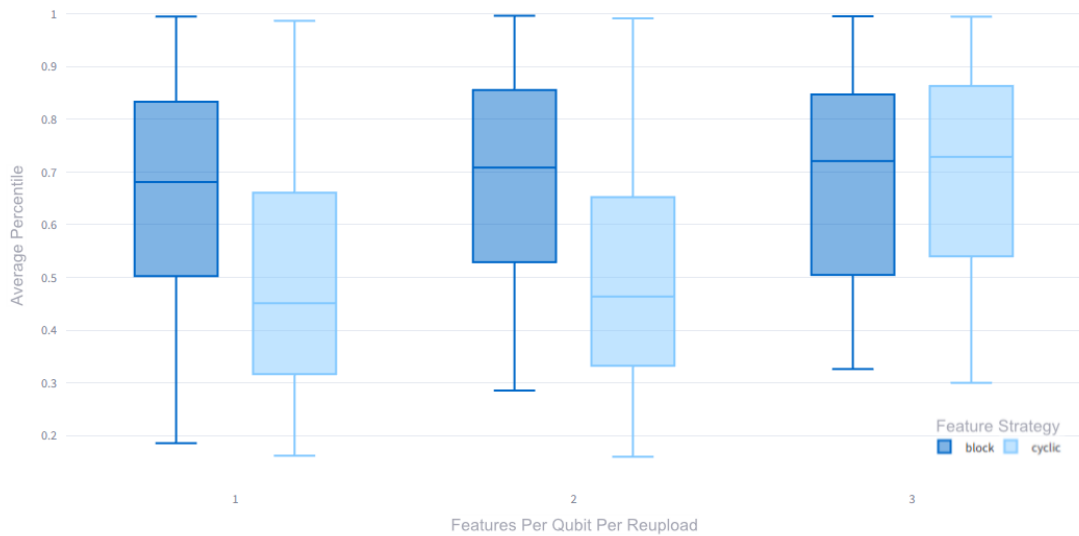


Figure 4.4: VQC performance by features per qubit and feature strategy.

4.3 Stage III: QuantumECOC Ensemble Analysis

This stage presents the ensemble-level part of the main analysis and directly evaluates the principal QuantumECOC contribution. It first examines whether the implemented framework could recover competitive multiclass performance through validation-weighted decoding, before assessing how learner count, architecture composition, information availability and feature allocation affected the outcome.

4.3.1 Predictive Performance Across Datasets

Figure 4.5 compares the distribution of five-run mean macro-F1 across the direct QuantumECOC configurations, with the two classical references. The median of each box represents the typical performance obtained when the underlying ensemble architecture is not tailored to the dataset, while the upper whisker gives the mean macro-F1 of the strongest evaluated configuration. The separation between these values therefore shows how much performance can be gained through appropriate ensemble design.

The strongest configurations provide a clearer indication of the potential of the ensemble framework. On Iris, the upper result slightly exceeded both classical references, while the optimal Heart Disease configuration substantially exceeded them. The strongest Contraceptive Method Choice ensemble matched Random Forest and exceeded the SVC. On Balance Scale, Image Segmentation and Steel Plates, the upper ensemble result exceeded one of the two classical models, although it remained below the stronger reference. Obesity and Yeast were the only datasets where the best direct ensemble remained below both classical models. Tailoring the underlying configuration therefore allowed the direct quantum ensemble to match or exceed at least one classical reference on six of the eight datasets.

These distributions show that the ECOC ensemble was capable of modelling the multiclass structure, but did not provide a uniform improvement across datasets. The width of several boxes is also important: Image Segmentation, Steel Plates and Obesity show that performance could change substantially according to the selected ensemble configuration. The next comparisons therefore examine which design choices were most consistently associated with the stronger direct results.

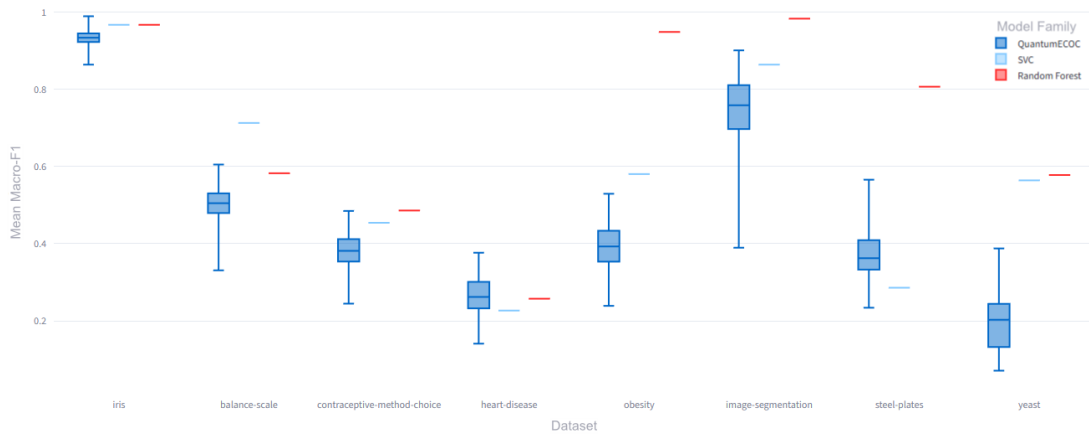


Figure 4.5: Distribution of test macro-averaged F1 score across the direct QuantumECOC configurations, with the SVC and Random Forest reference scores shown for each dataset.

4.3.2 Learner Count and Architecture Composition

Figure 4.6 separates the average cross-dataset performance by learner count and trainable-layer group. Increasing the ensemble from 6 to 10 and then 14 base learners shifted almost every layout group towards a higher median. The improvement was particularly clear at 14 learners, where the lower-ranked groups reached performance levels comparable with, or above, the strongest six-learner groups. Learner count was therefore the most consistent ensemble-level factor in this comparison.

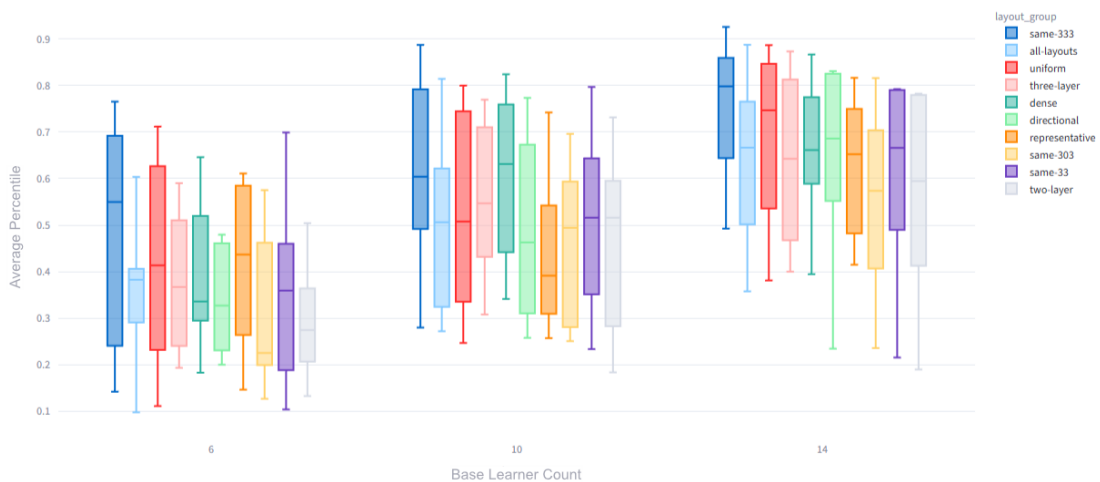


Figure 4.6: Distribution of average cross-dataset rank percentile by base-learner count and trainable-layer group.

Architecture composition still affected the result within each learner count. The homogeneous `same-333` group produced the highest median among the six- and fourteen-learner ensembles and remained competitive at ten learners. The dense group produced the highest median at ten learners, while the uniform group also performed strongly at fourteen. In contrast, combining all retained layouts did not consistently outperform the more restricted groups. Architectural variation therefore did not provide an automatic benefit, an ensemble of consistently strong layouts could outperform a broader mixture that also introduced weaker base designs.

The distributions nevertheless overlap, showing that learner count and layout group do not fully determine the result. Their interaction is more informative than either factor alone. Increasing the number of learners expanded the number of binary class partitions represented in the final

decision, but the benefit remained dependent on whether those learners used sufficiently effective circuit layouts.

4.3.3 Feature Allocation and Encoding Design

Figure 4.7 compares the remaining feature and encoding choices. Feature density produced the clearest separation, with the ordering of the distributions following the proportion of the original input made available to the learners. The stronger performance of the higher-density configurations therefore appears to reflect their broader feature coverage rather than an advantage associated with the parameter value itself. Lower-density ensembles introduced greater variation between learners, but did so by restricting the information available for each binary classification problem. The mixed configurations occupied an intermediate position, with top configurations achieving significantly higher performance, suggesting that this trade-off could be beneficial.

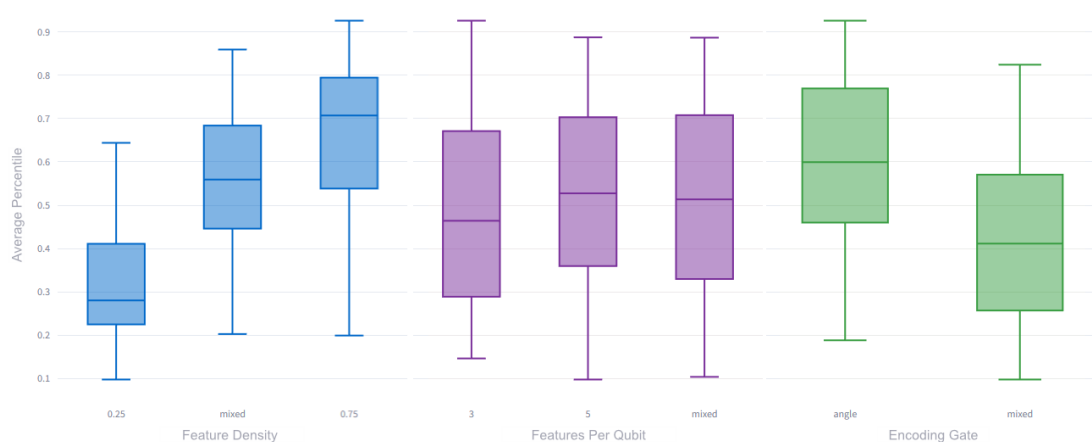


Figure 4.7: Distribution of average cross-dataset rank percentile by feature density, encoding-gate composition and features per qubit.

The difference between the features-per-qubit groups was smaller, with substantial overlap once enough encoding positions were available. This again indicates that increasing a single parameter is not independently responsible for stronger performance. The more important distinction was whether the combined configuration provided sufficient space to represent the available information without excessively restricting each learner.

Stage III Summary

Overall, the direct ensemble search shows that diversity was most effective after the information requirements of the base learners had been satisfied. Increasing learner count, varying feature assignments and combining circuit layouts could produce complementary predictions, but these benefits weakened when the individual circuits were given insufficient feature coverage or encoding capacity. Strong QuantumECOC configurations therefore balanced variation between learners with the need to preserve informative representations within each learner. These insights provide the fixed base configurations used in the subsequent comparisons of learned aggregation methods.

These relationships describe the evaluated two-qubit binary ECOC learners. Increasing the width of the base circuits or replacing the binary decomposition with multiclass learners may change the balance between individual learner capacity, task diversity and the number of components required by the ensemble.

4.4 Stage IV: Measured Aggregation Comparisons

This supporting comparison evaluates two learned aggregation extensions of a reduced set of ensemble configurations. Because the ECOC matrix, base architectures, learner count and feature assignments remain matched to previous configurations, changes in performance can be attributed to how the measured probability outputs are interpreted rather than to a different underlying ensemble. The results also establish the measured references required to evaluate CoherentECOC.

4.4.1 Meta-Training Procedure

- Compare the stratified 70:30 single-split and three-fold procedures where both were evaluated.
- Report predictive performance, repeated-run variation and total training cost under each procedure.
- State which procedure supplied the matched direct, stacked and coherent comparisons used in the final analysis.

4.4.2 Classical Stacking

- Compare the SVC stacker with the matched direct ensemble configurations.
- Report overall, per-dataset and class-level changes.
- Separate the cost of fitting the classical combiner from the base-ensemble training required by the selected procedure.

4.4.3 Measured Quantum Stacking

- Compare shortlisted meta-VQC designs with the classical SVC meta-learner.
- Analyse whether the meta-VQC architecture interacts with base ensemble characteristics.
- Report optimisation variability, meta-circuit size and training cost.

4.4.4 Stacking Candidate Reduction

- Identify which measured stackers progress to coherent construction.
- Retain several strong and structurally different candidates rather than only one top rank.
- Explain whether stacking gains concentrate on difficult or high-class-count datasets.

Research notes and evidence.

- Prior VQC composition work reports stacking gains substantially larger than bagging gains, providing a direct comparison for the measured stacking stage [16].

-
- Classical stacking research shows that different meta-learners can change class-level behaviour, supporting separate analysis of overall and minority-class performance [6].
 - Classical ensembles of single-qubit quantum classifiers have used logistic-regression meta-learning rather than simple voting, providing precedent for learned measured aggregation [13].

Results to insert.

- Matched direct-versus-stacked table.
- Classical-versus-quantum meta-learner ranking.
- Fold-training cost and convergence plots.

4.5 Stage V: CoherentECOC Innovation

This stage evaluates the dissertation's principal innovation. CoherentECOC replaces the measured interface between the pretrained base ensemble and meta-learner with one joint circuit, allowing aggregation to occur before final measurement. Its primary comparison is therefore with the matched direct and stacked conditions, while the resource analysis determines whether any benefit from pre-measurement interaction justifies the additional width, depth and optimisation cost.

4.5.1 Implementation and Trainability

- Report which coherent widths and candidate configurations were successfully constructed and trained.
- Report whether the full circuit was unfrozen for final end-to-end fine-tuning and analyse its effect where completed.
- Compare the 70:30 single-split and three-fold coherent training procedures where both were evaluated.
- Report convergence, repeated-run variability and computational limits.

4.5.2 Predictive Comparison

- Compare coherent models with their matched direct, classical-stack and measured-quantum-stack conditions.
- Analyse main-wire versus full-register meta designs and the retained meta-circuit variables.
- Identify dataset and ensemble characteristics associated with any coherent gain or loss.

4.5.3 Resource Comparison

- Report coherent width, trainable parameter count, logical depth and two-qubit operations.

- Compare independently trainable base parameters with the parameter count of equal-width monolithic VQCs.
- Discuss sequential training cost separately from the architectural possibility of parallel base training.

Research notes and evidence.

- Existing coherent ensemble proposals show that classifier contributions can be combined inside one quantum circuit, but their state-preparation assumptions and classifier models differ from pretrained VQC stacking [23, 26].
- Increasing coherent width and cross-register entanglement can worsen optimisation and routing overhead, so a lack of predictive gain must be interpreted alongside the additional resource cost [5, 20].
- The dissertation's coherent result should be compared primarily with matched measured stackers because previous VQC ensemble studies generally collapse each base learner before aggregation [13, 10].

Results to insert.

- Coherent feasibility and finalist table.
- Matched aggregation-family plots.
- Coherent circuit-resource table and diagrams.

4.6 Cross-Stage Ranking and Performance–Cost Analysis

- Present one global macro-averaged F1 score ranking across all completed test runs and model families.
- Reanalyse the ranking by stage, family, learner count, architecture group, dataset and resource tier.
- Compare mean, minimum, maximum and standard deviation across the five runs.
- Identify Pareto-efficient models and low-cost models above a defined performance threshold.
- Determine which variables remain influential after progressively more complex aggregation is introduced.

Results to insert.

- Full cross-stage rank table.
- macro-averaged F1 score against training-time and parameter-count plots.
- Summary of candidate survival between stages.

4.7 Summary by Research Question

- Provide one concise evidence-based answer for each research question.
- Reference the relevant tables and figures rather than repeating every numerical result.
- Distinguish statistically stable findings, provisional tendencies and unresolved questions.

Chapter 5: Discussion

5.1 VQC Component Analysis and Progressive Search Narrowing

- Interpret which VQC characteristics were consistently associated with performance, stability and training efficiency.
- Explain whether parameter count, reuploads and trainable-layer placement mattered independently or through interactions.
- Discuss whether sparse or reduced designs approached dense performance and why several varied candidates were retained.
- Evaluate whether progressive narrowing preserved scientifically useful diversity while avoiding wasteful downstream testing of poor models.
- Compare the findings with prior evidence on ansatz ordering, random feature subspaces and shallow circuits.

Research notes and evidence.

- Data-weaved quantum embedding circuits often outperform less effective layer orderings with the same nominal parameter count, supporting interpretation of layer placement rather than parameter count alone [22].
- Prior feature-subspace VQC results favour reduced feature ratios on some datasets but not universally, supporting dataset-dependent interpretation rather than one fixed optimum [17].
- Supervisor-provided Final Year Project (FYP) evidence is useful design guidance but should be distinguished from public peer-reviewed findings unless reproduced directly in this dissertation [17, 16].

5.2 QuantumECOC Ensemble Analysis and Measured Comparisons

- Interpret how ECOC reduced each multiclass problem to restricted binary tasks, shifting learner-level improvement towards circuit design and encoded information.
- Examine the task diversity created by the different binary class partitions alongside feature subsampling, learner count, bagging and architecture composition.

- Determine whether homogeneous strength or prediction diversity better explains ensemble gains.
- Discuss when direct aggregation was sufficient and when learned stacking added value.
- Compare classical and measured quantum meta-learners without attributing any gain to state preservation.
- Explain the substantial fold-training cost and whether the predictive improvement justified it.

Research notes and evidence.

- Prior VQC ensemble work identifies prediction disagreement as the strongest indicator of contribution and reports reduced cross-family variance after ensembling [16].
- Bagging is generally associated with variance reduction, while boosting addresses weak high-bias learners; no single ensemble strategy should be assumed universally preferable [10].
- Classical stacking studies often fix a small selected model count and identify principled model-count selection as future work, supporting explicit justification of the retained learner counts [6].

5.3 CoherentECOC Innovation and Measurement-Free Aggregation

- Determine whether the coherent model provided evidence that intermediate measurement acted as a practically relevant information bottleneck.
- Separate implementation feasibility from predictive superiority.
- Compare the single-split and fold-based training procedures and distinguish gains from broader meta-training coverage from their additional computational cost.
- Assess whether final end-to-end fine-tuning improved the coherent model after the base and meta stages had been trained separately.
- Compare main-wire and full-register processing and assess whether additional cross-register access justified its cost.
- Discuss whether any coherent result could instead be explained by parameter count, circuit capacity or training differences.
- State clearly when the evidence supports only comparable performance, reduced trainable parameters or modular training rather than improved classification.

Research notes and evidence.

- Measurement does not expose the complete quantum state, but this information-theoretic fact does not by itself establish that the discarded information is useful for the classification task [20].

- Early coherent ensemble proposals rely on idealised parameter superpositions or specialised indexed classifiers, so agreement or disagreement with them should be interpreted cautiously [23, 26].
- Measured ensemble studies leave open whether their resource-saving and robustness benefits persist when base circuits are connected coherently before final measurement [10].

5.4 Performance–Resource Trade-Offs and Practical Implications

- Interpret macro-averaged F1 score together with training time, parameters, circuit width, depth and two-qubit gate count.
- Distinguish total sequential compute from potential modular wall-clock parallelism.
- Discuss the effect of increasing learner count and coherent width on simulator feasibility.
- Identify which model families appear practically defensible under near-term resource constraints.

Research notes and evidence.

- Limited hardware connectivity can introduce quantum state swap gate (SWAP) operations that increase depth, latency and error exposure [20].
- Resource-saving QNN ensembles can reduce qubit requirements substantially, but the benefit depends on feature coverage and dataset structure [10].
- Full quantum advantage for Noisy Intermediate-Scale Quantum (NISQ) QML remains unresolved, so practical conclusions should be based on observed performance–resource trade-offs rather than theoretical possibility alone [8].

5.5 Relationship to Existing Literature

- Compare the VQC architecture findings with data re-uploading, ansatz-ordering and feature-subspace studies.
- Compare direct and stacked ensembles with measured quantum bagging, boosting and multi-chip approaches.
- Compare coherent findings with theoretical classifier superpositions and weighted coherent ensembles.
- Identify where the dissertation reproduces prior tendencies, contradicts them or extends them through matched aggregation comparisons.
- Avoid treating differences in datasets, classifier types, circuit access assumptions or noise models as direct like-for-like evidence.

5.6 Limitations and Threats to Validity

5.6.1 Internal and Conclusion Validity

- Discuss remaining sensitivity to parameter initialisation, finite repetitions and early-stopping choices.
- Explain how the fixed splits improve comparability but limit estimation across alternative sampled partitions.
- Discuss the progressive candidate-selection procedure and the risk that early ranking removes configurations that might interact favourably with later aggregation.
- State the statistical analyses actually completed and avoid claims unsupported by five-run uncertainty.

5.6.2 Construct Validity

- Discuss whether macro-averaged F1 score and the selected resource measures adequately represent useful model quality.
- Clarify that measured probabilities are a restricted classical representation, while coherent circuits do not provide a direct measurement of preserved information content.
- Distinguish equal-width from equal-capacity monolithic comparisons.

5.6.3 External Validity

- Limit generalisation to the selected UCI tabular multiclass datasets and implemented ansatz families.
- State that the Stage III relationships between learner count, feature allocation, architecture composition and diversity may change when the base learners use more qubits or learn multiclass tasks directly.
- Consider whether wider, higher-capacity learners could achieve comparable ensemble coverage with fewer components rather than assuming that the retained learner counts remain appropriate.

Research notes and evidence.

- Many prior quantum ensemble studies use small datasets, reduced features, ideal simulation or specialised state-access assumptions; their reported gains therefore do not establish broad deployment performance [27, 26].
- Persistent quantum memory and repeated access to stored intermediate states remain impractical, constraining how coherent stacking can be implemented beyond a single composed circuit [20].

Chapter 6: Conclusion and Future Work

6.1 Summary and Answers to the Research Questions

- Restate QuantumECOC as the principal contribution and summarise the multiclass functionality implemented by the framework.
- Summarise the main VQC and ensemble analysis, including information capacity, trainability, learner count, architecture composition and feature availability.
- Summarise the supporting direct, classical-stacked and measured-quantum comparisons.
- State whether the CoherentECOC innovation was feasible and whether it improved, matched or reduced predictive performance relative to measured alternatives.
- Provide one concise answer to each research question using only completed evidence.

6.2 Contributions

- State the deterministic QuantumECOC implementation as the principal contribution.
- State the CoherentECOC circuit-construction and staged-training method as the principal innovation.
- State the cross-dataset VQC and ensemble design investigation as the main empirical analysis.
- State the configurable ansatz and VQC implementation as an enabling contribution.
- State the matched classical and measured-quantum aggregation comparisons as supporting evaluation.
- State the performance–resource contribution across the implemented framework.
- Remove any planned contribution that was not completed or evaluated.

6.3 Limitations and Future Work

- Extend the dataset range and repeat evaluation across additional train–test splits.
- Explore selection based directly on prediction diversity and complementary errors.

-
- Investigate alternative ECOC matrices and learned class decompositions without increasing base-task complexity excessively.
 - Adapt the ensemble classes to bypass ECOC decomposition and construct ensembles from multiclass VQC base learners selected using the architecture-search findings.
 - Use the framework's existing support for wider base learners to determine whether additional circuit and encoding capacity improves predictive performance while reducing the number of learners required.
 - Repeat the Stage III ensemble-design analysis for wider and multiclass base learners because their capacity and task formulation may change the observed relationships between learner quality, diversity, feature allocation and learner count.
 - Develop topology-aware and adaptive coherent meta-circuits.
 - Explore direct diagnostics of state similarity, mutual information or phase-sensitive relationships within coherent registers.
 - Evaluate larger coherent ensembles using tensor-network, distributed or approximate simulation.
 - Expand noise-aware training, mitigation and repeated real-hardware evaluation.
 - Compare additional capacity-matched monolithic and hybrid models.
 - Release and formally evaluate the architecture-search and result-analysis tooling where appropriate.

Research notes and evidence.

- The practical crossover at which QML becomes preferable to classical approaches remains unresolved and should continue to be evaluated through controlled performance and resource comparisons [4].
- Prior VQC ensemble work identifies quantum meta-learning and coherent state-preserving aggregation as natural extensions, but large-scale hardware validation remains open [16, 10].
- Quantum hardware should be treated as a specialised and evolving computational resource, so future work should prioritise task-specific evidence rather than universal quantum-advantage claims [20].

6.4 Final Conclusion

- Provide a concise final statement on what the study establishes through the QuantumECOC contribution, its VQC and ensemble analysis, and the CoherentECOC innovation.
- State the practical significance of the best-performing and most resource-efficient configurations.
- End with the strongest defensible conclusion rather than a speculative claim.

Appendix A: Dataset Details

- Data sources and licences.
- Original and cleaned dimensions.
- Feature definitions and class distributions.
- Dataset-specific preprocessing.

Appendix B: **VQC** and **Ansatz** Configurations

- Layer specifications.
- Final template JavaScript Object Notation (JSON) definitions.
- Parameter and gate counts.
- Representative circuit diagrams.

Appendix C: **Experimental Search Spaces**

- VQC search grid.
- Ensemble composition grid.
- Meta-circuit grid.
- Final reduced stress-test configurations.

Appendix D: **Additional Results**

- Full per-dataset tables.
- Classification reports.
- Confusion matrices.
- Loss curves.
- Failed-run summaries.

Appendix E: Circuit and Resource Details

- Individual VQC circuit.
- Direct QuantumECOC information flow.
- Re-encoding meta-VQC.
- Measurement-free coherent circuit.
- Logical resource tables.

Appendix F: Computational Environment and Reproducibility

- Software versions.
- High-Performance Computing (HPC) resources and Slurm scripts.
- Commands for job generation, execution and aggregation.
- Result schemas and dashboard instructions.

Acknowledgements

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